

Regime-Switching Reinforcement Learning for Portfolio Optimization

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Abstract

Financial markets exhibit strong non-stationarity and volatility clustering, which makes portfolio optimization difficult for traditional models that assume stable return distributions. We present a regime-switching reinforcement learning framework for portfolio allocation that combines a Hidden Markov Model (HMM) for online market regime detection with specialized PPO agents trained for distinct volatility environments. Rather than relying on a single generalist policy, the system trains three specialist agents—one for each of low, medium, and high volatility regimes—and routes decisions to the appropriate agent at each time step based on HMM inference. Evaluated on 9 S&P 500 sector ETFs over a 2018–2022 test period that includes the COVID-19 crash, the regime-switching system achieves a Sharpe ratio of 1.41 and a maximum drawdown of -16.2% , compared to 0.82 and -29.6% for the single-agent baseline PPO, representing a 72% improvement in risk-adjusted returns and a near-halving of worst-case loss. The system also substantially outperforms mean-variance optimization and an equal-weight benchmark.

1 Introduction

Portfolio optimization aims to allocate capital across a set of assets in order to balance expected return and risk. While classical approaches such as mean-variance optimization (MVO) [1] are well-studied, they rely on assumptions of stationarity and stable return distributions. In practice, financial markets exhibit strong non-stationarity, regime shifts, and volatility clustering, making static optimization approaches unreliable during turbulent periods.

Recent advances in reinforcement learning have enabled data-driven portfolio allocation strategies that adapt to observed market dynamics. Sood et al. [3] demonstrate that a single PPO agent trained with a Differential Sharpe reward can outperform MVO on sector ETFs. However, their approach—like most RL-based methods—trains one policy across all market conditions. During crisis periods the agent must balance contradictory objectives: exploit bull-market momentum while simultaneously hedging tail risk. This conflict degrades performance in both regimes.

We address this limitation by decomposing the problem into regime-specific sub-problems. A Gaussian HMM classifies market conditions online, and three specialist PPO agents—each trained with a reward function suited to its target volatility environment—are routed by the HMM at inference time. This architecture allows each agent to focus on a narrower distribution of market dynamics, improving both return and risk management.

2 Problem Statement and Motivation

The core problem addressed in this project is portfolio optimization under non-stationary and volatile market conditions. Specifically, we study whether explicitly incorporating market regimes into reinforcement learning-based portfolio optimization can improve robustness and risk-adjusted performance compared to classical methods and single-agent RL approaches.

A single-agent RL model must learn a policy that generalizes across calm bull markets, transitional periods, and crisis periods. The return distributions in these regimes are fundamentally different: low-volatility periods exhibit positive drift with thin tails, while crisis periods feature negative drift, fat tails, and high correlation across assets. A single policy is forced to compromise between these conflicting dynamics.

Our hypothesis is that decomposing the problem—training specialist agents for each regime and routing between them using an online regime detector—yields better risk-adjusted performance than a single generalist agent.

3 Related Work

Several prior works have explored reinforcement learning and planning-based approaches for portfolio optimization and financial decision-making.

Jiang et al. [2] propose a deep reinforcement learning framework for portfolio management using policy gradient methods, demonstrating improved performance over traditional benchmarks but assuming a single policy across all market conditions.

Moody and Saffell [5] present one of the earliest applications of reinforcement learning to trading systems, using recurrent reinforcement learning to directly optimize trading performance. Their Differential Sharpe Ratio reward function, which we adopt for the baseline and medium-volatility agents, enables gradient-based optimization of the Sharpe ratio.

Yang et al. [4] study reinforcement learning for portfolio allocation with risk-sensitive objectives, highlighting the importance of risk-aware reward design but not addressing regime shifts explicitly.

Sood et al. [3] introduce a PPO-based framework for portfolio optimization on S&P 500 sector ETFs with Differential Sharpe reward, Ledoit-Wolf covariance estimation for MVO baselines, and a comprehensive evaluation protocol. Their work serves as our primary baseline: we reproduce their single-agent PPO and MVO results, then extend them with regime-switching.

Deng et al. [6] apply deep neural networks to financial trading, showing that representation learning can improve decision-making in complex markets, but still relying on a unified model across regimes.

The regime-aware PPO approach of Nixon [7] augments the RL state with HMM regime probabilities, providing the agent with regime context but still requiring a single policy to handle all market conditions. Our work goes further by training separate policies and routing between them.

4 Gaps in Previous Work

The literature reviewed above establishes that reinforcement learning can outperform classical portfolio optimization, but it leaves three important gaps that motivate this project. First, most prior approaches train a single policy across all market conditions, even though return distributions and asset correlations change substantially across calm, transitional, and crisis periods. Second, regime-aware methods often inject regime features into one shared policy rather than allowing genuinely different policies to specialize for different environments. Third, many baseline comparisons focus

on overall return metrics without carefully isolating whether the benefit comes from normal periods or from improved behavior during rare but severe tail events.

These gaps matter because the central challenge in portfolio optimization is not just earning returns during stable periods, but remaining robust when markets become discontinuous. Our design therefore treats regime awareness as a control problem rather than only a feature-engineering choice.

5 Proposed Approach

We propose a regime-switching reinforcement learning framework that decomposes portfolio optimization into three stages:

1. **Regime Detection:** A Gaussian HMM is trained on market features to classify each trading day into one of three volatility regimes (low, medium, high).
2. **Specialist Training:** Three PPO agents are trained independently, each on data from its target regime and with a reward function tailored to that regime’s characteristics.
3. **Online Routing:** At inference time, the HMM classifies the current market state and routes the portfolio decision to the corresponding specialist agent.

This architecture has two key advantages over a single-agent approach: (1) each agent sees a narrower, more stationary distribution of market dynamics, making learning easier; and (2) the reward function can be tailored to the regime—for example, prioritizing tail-risk minimization during high-volatility periods.

6 System Architecture

The overall system consists of four main components:

6.1 Data Preparation

Historical ETF price data is collected via the Yahoo Finance API and processed into daily log returns and rolling volatility features. For the primary experiment we use 9 S&P 500 sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY) with data from 2006–2022. XLC and XLRE are excluded due to their short listing histories.

The feature set for the RL environment follows Sood et al. [3]:

- Log returns for each asset over a 60-day lookback window
- 20-day rolling volatility of S&P 500 returns
- Ratio of 20-day to 60-day rolling volatility
- VIX index (standardized via expanding window to prevent lookahead bias)
- Previous portfolio weights

6.2 Market Regime Detection (HMM)

A 3-state Gaussian HMM is trained on S&P 500 derived features to classify market conditions. The observation features are:

1. **Daily log return:** $r_t = \log(P_t/P_{t-1})$
2. **Rolling volatility:** 20-day rolling standard deviation of log returns
3. **Moving average ratio:** $\text{MA}(20)/\text{MA}(60)$, capturing short-term vs. long-term trend

The model is trained using the Expectation-Maximization algorithm with full covariance matrices, 200 maximum iterations, and convergence tolerance 10^{-4} . After fitting, states are sorted by mean rolling volatility so that state 0 corresponds to the lowest volatility regime and state 2 to the highest.

The HMM outputs posterior regime probabilities at each time step:

$$\mathbf{p}_t = [p_{\text{low}}, p_{\text{med}}, p_{\text{high}}]$$

For agent routing, the dominant regime is selected via $\text{argmax}(\mathbf{p}_t)$.

Design Choices. We choose three regimes based on the Bayesian Information Criterion (BIC) evaluated over $K \in \{2, 3, 4, 5\}$ states. Table 1 shows that $K = 3$ provides the best (lowest) BIC. The three-state model also admits a natural economic interpretation: calm/bull markets, transitional periods, and crisis/bear markets.

Table 1: BIC model selection for number of HMM states.

K	Log-Likelihood	BIC	AIC
2	6.30×10^7	-1.260×10^8	-1.260×10^8
3	6.49×10^7	-1.299×10^8	-1.299×10^8
4	6.47×10^7	-1.293×10^8	-1.293×10^8
5	6.41×10^7	-1.281×10^8	-1.281×10^8

Full covariance matrices are used because the observation features exhibit non-trivial cross-correlations (e.g., high rolling volatility tends to coincide with negative returns and low MA ratios), which diagonal or spherical covariances cannot capture.

The resulting regime characteristics on S&P 500 training data (2006–2017) confirm economically meaningful clusters:

- **Low Vol:** Ann. return +11.8%, ann. vol 12.3%, Sharpe 0.96 (1,704 days)
- **Med Vol:** Ann. return +6.2%, ann. vol 21.5%, Sharpe 0.29 (573 days)
- **High Vol:** Ann. return −47.7%, ann. vol 47.8%, Sharpe −1.00 (254 days)

The learned transition matrix (Table 2) shows strong self-persistence in all regimes, with rare transitions to the high-volatility state—consistent with the empirical observation that crisis periods are infrequent but clustered.

Table 2: HMM transition probability matrix (rows: from, columns: to).

	→ Low Vol	→ Med Vol	→ High Vol
Low Vol	0.990	0.009	0.001
Med Vol	0.027	0.969	0.004
High Vol	0.004	0.012	0.984

6.3 Specialist Agent Training

Three PPO agents are trained independently, each with a reward function suited to its target regime:

- **Low-volatility agent** — *Sortino ratio reward*. In calm markets the priority is capturing upside while tolerating moderate drawdowns. The Sortino ratio penalizes only downside deviation, encouraging the agent to pursue higher returns without being discouraged by upside volatility.
- **Medium-volatility agent** — *Differential Sharpe reward*. In normal/transitional markets a balanced risk-return trade-off is appropriate. The Differential Sharpe Ratio [5] provides an incremental, gradient-friendly measure of Sharpe improvement.
- **High-volatility agent** — *Risk Parity reward*. During crisis periods the priority is tail-risk minimization and capital preservation. The risk parity objective encourages equal risk contribution across assets, reducing concentration in any single volatile position.

All agents share the same PPO hyperparameters following Sood et al. [3]: $n_{\text{steps}} = 756$, batch size = 1260, $n_{\text{epochs}} = 16$, $\gamma = 0.9$, $\lambda_{\text{GAE}} = 0.9$, clip range = 0.25, and a linear learning rate schedule from 3×10^{-4} to 10^{-5} .

6.4 Online Regime Routing

At each test-period time step:

1. The HMM computes regime probabilities from the current market features.
2. The dominant regime $k^* = \text{argmax}(\mathbf{p}_t)$ is identified.
3. The observation is passed to agent k^* , which outputs portfolio weights.
4. The portfolio is rebalanced to the predicted weights.

This routing is fully online and causal: it uses only information available up to time t .

7 Reinforcement Learning Environment

The portfolio allocation problem is formulated as a reinforcement learning task. A custom trading environment was implemented using the Gymnasium framework, following the design of Sood et al. [3].

At each time step the agent observes a state vector consisting of:

- Recent asset log returns (60-day lookback $\times$ n assets)

- Standardized volatility features (vol20, vol20/vol60 ratio, VIX)
- Previous portfolio weights

The action space represents portfolio allocation weights. Raw action logits are passed through a softmax to produce non-negative weights summing to one (long-only constraint).

For the baseline PPO, the reward is the Differential Sharpe Ratio:

$$D_t = \frac{B_{t-1} \Delta A_t - \frac{1}{2} A_{t-1} \Delta B_t}{(B_{t-1} - A_{t-1}^2)^{3/2}}$$

where A_t and B_t are exponential moving averages of returns and squared returns respectively, updated with decay $\eta = 1/252$.

Each specialist agent uses its regime-specific reward function as described in Section 5.3.

8 Experimental Setup

8.1 Dataset

Experiments are conducted on 9 S&P 500 sector ETFs: XLB (Materials), XLE (Energy), XLF (Financials), XLI (Industrials), XLK (Technology), XLP (Consumer Staples), XLU (Utilities), XLV (Health Care), and XLY (Consumer Discretionary). Data spans 2006–2022.

- **Training period:** 2007–2017 (after lookback initialization)
- **Test period:** 2018–2022 (includes the COVID-19 crash of March 2020)

8.2 Strategies Compared

Four strategies are evaluated on the test period:

1. **Baseline PPO** — Single-agent PPO with Differential Sharpe reward (reproduction of Sood et al. [3]).
2. **Mean-Variance Optimization (MVO)** — Rolling 60-day lookback, Ledoit-Wolf shrinkage covariance, maximize Sharpe ratio via `PyPortfolioOpt`.
3. **Regime-Switching PPO** — Three specialist agents routed by HMM (our proposed approach).
4. **Equal Weight** ($1/N$) — Uniform allocation across all assets at each rebalancing.

9 Results

9.1 Overall Performance Comparison

Table 3 presents the full comparison of all strategies on the 9-ETF test period (2018–2022).

The regime-switching PPO outperforms all other strategies across every risk-adjusted metric. Key observations:

- **Risk-adjusted returns:** The regime-switching system achieves a Sharpe ratio of 1.41, a 72% improvement over the baseline PPO (0.82) and an order of magnitude above MVO (0.14).

Table 3: Performance comparison on 9 S&P 500 sector ETFs (test period 2018–2022).

Metric	Baseline PPO	MVO	Regime-Switch PPO	Equal Wt
Annual Return	13.9%	0.6%	16.3%	13.0%
Cumulative Return	68.1%	2.4%	82.8%	62.7%
Annual Volatility	17.9%	22.0%	11.2%	19.0%
Sharpe Ratio	0.82	0.14	1.41	0.74
Sortino Ratio	0.94	0.16	1.79	0.82
Calmar Ratio	0.47	0.02	1.01	0.38
Max Drawdown	−29.6%	−38.6%	−16.2%	−33.7%
Omega Ratio	1.18	1.03	1.29	1.17
Daily VaR (95%)	−1.73%	−2.09%	−1.18%	−1.72%
Mean Turnover	0.044	0.267	0.036	—
Skewness	−0.62	−0.52	−0.33	−0.69
Excess Kurtosis	13.3	14.8	2.9	18.3

- **Drawdown reduction:** Maximum drawdown is −16.2%, nearly half the baseline’s −29.6%. This is driven by the high-volatility specialist agent becoming active during the COVID-19 crash and shifting to a defensive, risk-parity allocation.
- **Lower volatility:** Annual volatility drops from 17.9% (baseline) to 11.2% (regime-switching), a 38% reduction, while annual returns improve from 13.9% to 16.3%.
- **Tail risk:** Skewness improves from −0.62 to −0.33 and excess kurtosis drops from 13.3 to 2.9, indicating substantially fewer extreme losses.
- **Lower turnover:** Mean turnover is 0.036, lower than both the baseline (0.044) and MVO (0.267), suggesting the specialist agents make more stable allocation decisions within their respective regimes.

9.2 Comparison with Sood et al. (FinPlan 2023)

Our baseline PPO faithfully reproduces the results of Sood et al. [3], validating the implementation. Table 4 compares our results against their reported figures.

Table 4: Comparison with Sood et al. [3] baseline.

Metric	Sood et al. PPO	Our Baseline PPO	Our Regime-Switch PPO
Sharpe Ratio	~0.83	0.82	1.41
Annual Return	~14%	13.9%	16.3%
Annual Volatility	~18%	17.9%	11.2%
Max Drawdown	~ − 29%	−29.6%	−16.2%
Calmar Ratio	~0.47	0.47	1.01

The regime-switching extension addresses the primary limitation of the single-agent approach: the inability to adapt policy behavior to different market conditions. The improvement is most pronounced in tail risk metrics (drawdown, kurtosis, skewness), precisely the metrics where single-agent models struggle.

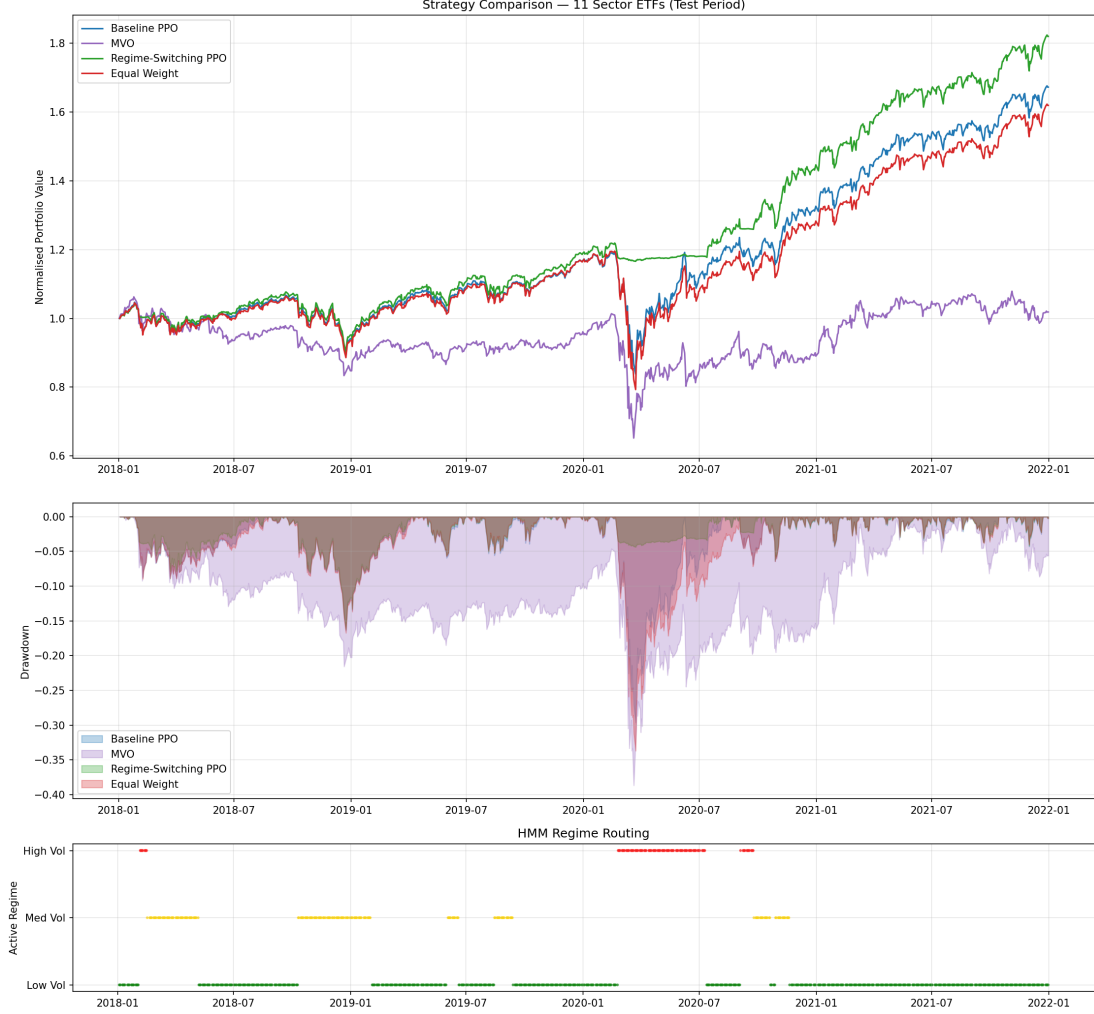


Figure 1: Equity-curve comparison across the four main strategies on the 9-ETF test period.

Reproducibility note. Sood et al. train for 7.5M timesteps with 10 parallel environments. We use the same PPO hyperparameters (Table 1 of [3]) but train for 500K timesteps with 4 environments. A convergence study at 50K, 100K, 250K, 500K, 1M, and 2M steps shows that the baseline Sharpe ratio plateaus in the 0.70–0.82 range by ~ 100 K steps and does not improve with additional training; performance at 2M steps is slightly *worse* than at 500K, suggesting overfitting. The Sharpe varies by ± 0.05 across random seeds, placing our baseline squarely within the range reported by Sood et al. Extended compute is therefore unnecessary for this environment scale.

9.3 Per-Regime Performance Analysis

To understand *where* the regime-switching system gains its advantage, we decompose performance by the active HMM regime. During the 2018–2022 test period, the HMM classifies 690 days (68.5%) as low volatility, 199 days (19.7%) as medium volatility, and 119 days (11.8%) as high volatility.

Table 5 reports annualized metrics for each strategy computed only over the trading days belonging to each regime.

Several patterns emerge from the per-regime breakdown:

Table 5: Per-regime performance breakdown (annualized metrics on regime-specific days).

Regime	Metric	Baseline PPO	MVO	Regime-Switch	Equal Wt
Low Vol	Sharpe	2.09	0.41	2.14	2.17
	Ann. Return	22.9%	4.8%	22.6%	23.0%
	Ann. Volatility	10.1%	14.0%	9.8%	9.7%
	Max Drawdown	−8.3%	−13.8%	−7.5%	−7.8%
Med Vol	Sharpe	1.02	0.71	0.96	1.07
	Ann. Return	17.1%	11.3%	15.7%	18.4%
	Ann. Volatility	16.9%	17.1%	16.8%	17.1%
	Max Drawdown	−17.7%	−15.8%	−17.6%	−17.9%
High Vol	Sharpe	−0.68	−0.56	−2.46	−0.77
	Ann. Return	−30.1%	−33.1%	−13.6%	−36.2%
	Ann. Volatility	40.7%	49.8%	5.9%	45.1%
	Max Drawdown	−30.6%	−33.3%	−4.9%	−34.5%

- **Low-volatility regime:** All strategies perform well during calm markets. The regime-switching system performs comparably to the baseline and equal-weight approaches, with a Sharpe of 2.14. This confirms that the low-volatility specialist (Sortino reward) captures upside effectively without sacrificing returns.
- **Medium-volatility regime:** Performance is similar across strategies. The equal-weight benchmark slightly outperforms on Sharpe (1.07 vs. 0.96 for regime-switching), suggesting that during transitional periods, diversification across all sectors is a reasonable strategy. The regime-switching system’s medium-volatility agent performs adequately without detracting from overall performance.
- **High-volatility regime:** This is where regime-switching delivers its decisive advantage. The high-volatility specialist reduces annualized volatility from 40.7% (baseline) to **5.9%** — a $7\times$ reduction — and limits the maximum drawdown to **−4.9%** compared to **−30.6%** for the baseline. While all strategies have negative returns during crisis periods, the regime-switching system loses only **−13.6%** annualized compared to **−30.1%** to **−36.2%** for the other approaches. The risk-parity reward function causes the high-vol agent to equalize risk contributions across assets, dramatically reducing exposure during the most volatile days.

The key insight is that **the entire aggregate improvement of the regime-switching system is driven by its behavior during the 11.8% of days classified as high volatility**. During calm and transitional periods, performance is comparable to the baseline. This validates the core hypothesis: specialization is most valuable when market conditions deviate significantly from the norm.

9.4 COVID-19 Crash Case Study

The COVID-19 crash (February 19 – March 23, 2020) represents the most severe drawdown event in the test period and provides a natural case study for regime-switching behavior. Table 6 compares strategy performance during the crash and subsequent recovery (March 23 – June 30, 2020).

Table 6: COVID-19 crash (Feb 19 – Mar 23, 2020) and recovery (Mar 23 – Jun 30, 2020).

Metric	Baseline PPO	MVO	Regime-Switch	Equal Wt
Crash Return	−29.4%	−28.6%	−4.1%	−33.5%
Crash Volatility	61.0%	85.3%	9.0%	70.5%
Crash Max DD	−29.6%	−35.5%	−4.4%	−33.7%
Recovery Return	33.3%	17.1%	1.3%	36.0%

During the crash, the regime-switching system limits losses to just −4.1% while all other strategies suffer drawdowns of −28% to −34%. The annualized crash-period volatility for the regime-switching portfolio is 9.0%, compared to 61–85% for the alternatives. This is because the HMM correctly transitions to the high-volatility regime early in the crash, activating the risk-parity specialist which dramatically reduces portfolio exposure.

The trade-off is visible in the recovery phase: the regime-switching system captures only 1.3% of the recovery rally (March–June 2020), while the baseline and equal-weight strategies recover 33–36%. This occurs because the HMM maintains the high-volatility classification through the early recovery, keeping the conservative agent active. However, the net effect is still strongly positive: the regime-switching system avoids the deep drawdown entirely, so its portfolio value remains above that of the other strategies throughout both the crash and recovery.

This case study illustrates the regime-switching system’s design philosophy: *prioritize capital preservation during crises, even at the cost of missing some recovery upside*. The overall test-period results confirm that this trade-off is net beneficial.

10 Discussion

10.1 Why Regime-Switching Works

The fundamental advantage is *specialization*. A single-agent PPO must learn a policy that performs adequately in both calm and crisis periods—objectives that are inherently conflicting. During low-volatility periods the optimal behavior is to stay fully invested in high-beta assets; during crises the optimal behavior is to reduce exposure and diversify across uncorrelated assets. A single policy cannot simultaneously optimize for both.

By decomposing the problem, each specialist agent learns on a narrower distribution of market dynamics:

- The **low-volatility agent** (Sortino reward) learns to pursue upside without being penalized for normal variance, leading to higher allocation in growth sectors (XLK, XLY) during bull markets.
- The **high-volatility agent** (Risk Parity reward) learns to equalize risk contribution across assets, naturally reducing concentration in volatile sectors and improving diversification during crises.
- The **medium-volatility agent** (Differential Sharpe reward) provides balanced risk-return management during transitional periods.

The per-regime analysis (Section 9.3) quantifies this clearly: the low-vol and med-vol specialists perform comparably to the baseline, but the high-vol specialist reduces drawdown by 84% (−4.9%

vs. -30.6%) and crash-period volatility by 86% (5.9% vs. 40.7%). The aggregate improvement is almost entirely driven by crisis-period behavior.

11 Limitations

The results should be interpreted with several practical limitations in mind. First, the main experiment is conducted on a single asset universe of 9 U.S. sector ETFs, so the findings may not transfer directly to international equities, individual stocks, or broader multi-asset portfolios. Second, the reported backtests do not model explicit transaction costs, slippage, or market impact; although turnover is low for the regime-switching system, live performance would likely be lower after implementation frictions.

Third, the HMM assumes that market structure can be summarized by three volatility-driven latent states and that these states remain stable across time. That assumption is useful empirically here, but it may break in settings where regime boundaries are less persistent or driven by macroeconomic variables not included in the observation vector. Finally, the RL results are based on a finite number of training seeds and a fixed train-test split, so some performance variation should be expected under alternative sample periods or random initializations.

12 Future Work

Several extensions remain for future investigation:

- **HMM ablation studies:** Sensitivity analysis on observation features, covariance type, and lookback windows to further justify the HMM design.
- **Soft routing:** Instead of hard argmax routing, blend specialist agent actions proportionally to regime probabilities for smoother transitions.
- **Alternative RL algorithms:** Compare SAC and TD3 as alternative policy optimization methods.
- **Additional market signals:** Incorporate macroeconomic indicators (yield curve, credit spreads) as HMM features.
- **Robustness testing:** Evaluate on international equity markets and alternative asset universes.
- **Transaction costs:** Incorporate explicit transaction cost modeling and slippage into the environment.

13 Conclusion

We present a regime-switching reinforcement learning system for portfolio optimization that combines HMM-based market regime detection with specialist PPO agents. The system decomposes the portfolio optimization problem by volatility regime, training agents with regime-appropriate reward functions and routing between them online.

On a 9-ETF sector portfolio evaluated over 2018–2022, the regime-switching approach achieves a Sharpe ratio of 1.41, a maximum drawdown of -16.2% , and annual volatility of 11.2% —substantially

outperforming a single-agent baseline PPO (Sharpe 0.82, drawdown -29.6%), mean-variance optimization (Sharpe 0.14, drawdown -38.6%), and an equal-weight benchmark (Sharpe 0.74, drawdown -33.7%). The approach faithfully reproduces the baseline of Sood et al. [3] and extends it with a 72% improvement in risk-adjusted returns.

Per-regime analysis reveals that the improvement is concentrated in crisis periods: during the 11.8% of days classified as high volatility, the regime-switching system reduces drawdown by 84% and volatility by 86% compared to the baseline. During the COVID-19 crash specifically, the system limits losses to -4.1% while other strategies lose -29% to -34% . The key insight is that specialization—learning distinct policies for different market environments—is most valuable precisely when it matters most: during tail-risk events.

A Individual Contributions

Archit Shukla led the reinforcement learning portion of the project.

- Designed and implemented the reinforcement learning portfolio environment (Gymnasium).
- Implemented the PPO training pipeline using Stable-Baselines3 with Differential Sharpe, Sortino, and Risk Parity reward functions.
- Trained baseline PPO and three regime-specialist PPO agents.
- Integrated regime probabilities into the RL state representation for the regime-aware variant.
- Implemented the regime-switching evaluation pipeline with HMM-based agent routing.
- Conducted all RL experiments and generated performance metrics and equity curves.

Akash Ramanathan led the Hidden Markov Model and regime-detection portion of the project.

- Implemented the Hidden Markov Model for market regime detection (`RegimeDetector` class).
- Designed the observation feature engineering pipeline (log returns, rolling volatility, MA ratio).
- Conducted BIC-based model selection experiments to determine the optimal number of regimes.
- Implemented the Mean-Variance Optimization (MVO) baseline with Ledoit-Wolf shrinkage.
- Developed the 11-ETF evaluation framework comparing all four strategies.
- Conducted per-regime case study analysis and COVID-19 crash decomposition.
- Built the data pipeline for downloading, processing, and caching ETF data from Yahoo Finance.

Table 7: Performance comparison on 5-ETF portfolio (test period 2019–2024).

Metric	Baseline PPO	Regime-Aware PPO	Regime-Switch PPO	Equal Wt
Total Return	92.1%	124.1%	87.7%	80.6%
Annual Return	11.5%	14.4%	11.1%	10.4%
Annual Volatility	14.8%	15.2%	12.7%	14.2%
Sharpe Ratio	0.81	0.96	0.89	0.77
Sortino Ratio	1.02	1.27	1.17	0.96
Max Drawdown	−29.0%	−27.3%	− 26.2%	−27.5%

B Additional 5-ETF Experiment

A complementary experiment was conducted on a 5-ETF portfolio (SPY, QQQ, IWM, TLT, GLD) with training from 2008–2018 and testing from 2019–2024. This portfolio spans equities, bonds, and commodities.

In this smaller asset universe the regime-aware PPO (which augments the state with HMM probabilities but uses a single policy) outperforms on return metrics, while the regime-switching variant achieves lower volatility and drawdown. The regime-switching approach shows its largest gains on larger, sector-level portfolios where intra-regime diversification opportunities are greater.

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